Sultanate of Oman
Ministry of Health
The Royal Hospital
Department of Child Health

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Title: Intravenous Fluid Therapy

Rationale:-
Maintenance intravenous fluids (IVF) are used to provide critical supportive care for children who are acutely ill but it is not a completely benign intervention.
Appropriate selection of type and rate of IVF could reduce IVF related adverse affects including hyponatremia.

Objective:-
Standardize care of pediatric patients who require maintenance IVF.

Indication of IVF
1- Maintenance IVF therapy to replace estimated normal physiologic urine output and insensible looses in patient with reduced or no oral intake.
2- Bolus fluid therapy to expand the circulating volume in children with hypovolemia and shock.
3- Replacement fluid therapy to replace abnormal losses from GI tract and other body cavities.

General Principles:-
- IVF should be considered as a medication and only given when needed and the child is unable to take orally such as NPO status.
- Most hospitalized children requiring IVF should be considered at risk of non-physiological SIADH.
- Oral fluid must be included in the estimation of total fluid intake.
- Infants and young children have limited glycogen stores, therefore, saline solution with added dextrose are required to prevent hypoglycemia and ketosis.

Inclusion Criteria:-
General pediatric patients
Exclusion criteria:-
- Renal disease /renal dysfunction
- endocrine disorder causing electrolyte abnormalities
- Neurosurgery or brain injury
- Liver failure/hepatic dysfunction
- Severe malnutrition
- Severe cardiac diseases

Identify risk of hyponatremia:-

- Pulmonary disease (pneumonia)
- Pain
- CNS disease
- Recent surgery
- Gastroenteritis and nausea/vomiting in general

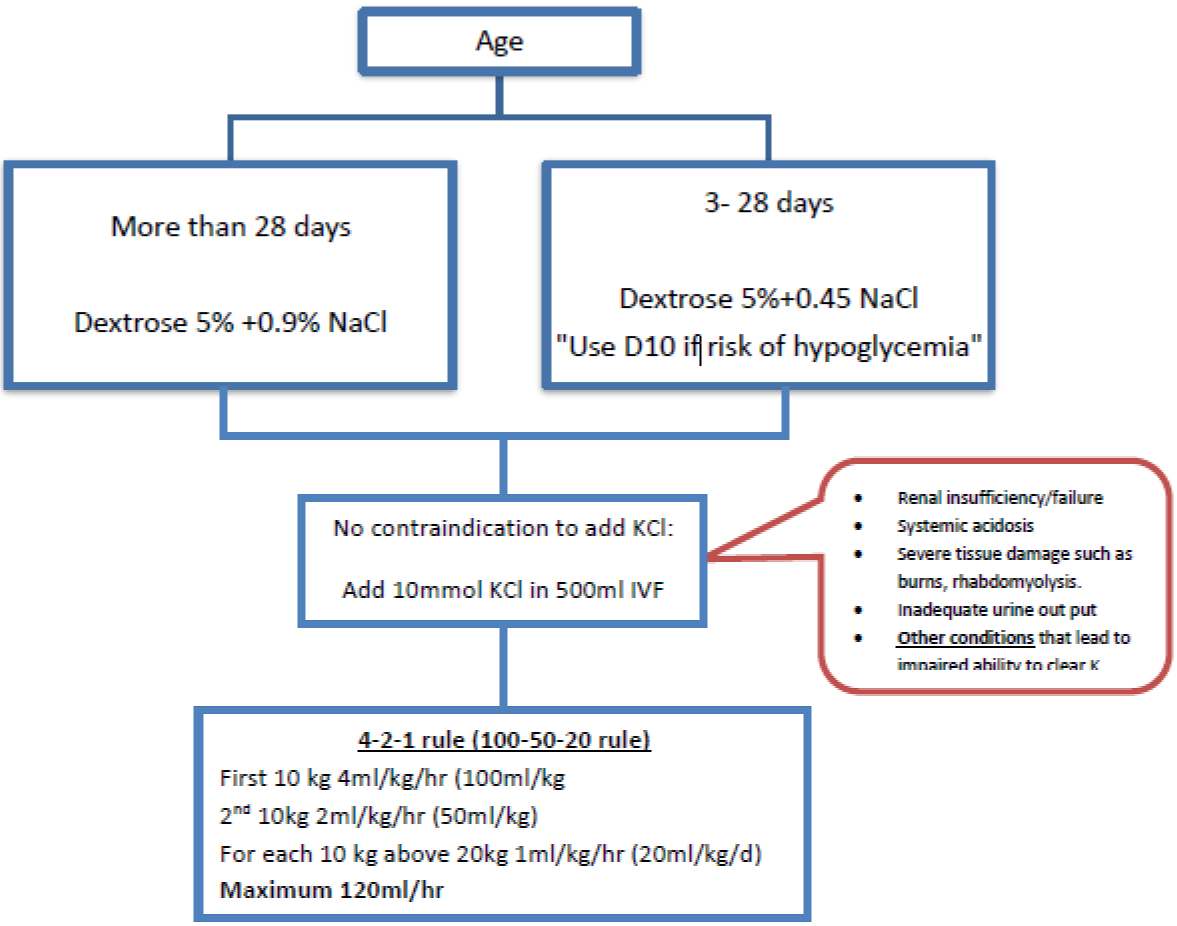
ELECTION OF MAINTENANCE IVF

Prior to starting fluids:

Send electrolyte and renal function.

Revise IVF prescription if needed after lab result.

If Na >145 change IVF to 0.45 NaCl (review cause of hypernatremia).



- Consider NGT feed if oral feed can't be given /tolerate due to respiratory issue.
- Monitoring: IVF should be treated like a medication with appropriate monitoring.

o If there is risk of SIADH, assess patient status and requirements of routine maintenance IVF.
o I/O charting (on going losses).
o Daily weight.
o Sign/symptoms of fluid retention (at least daily).
o Electrolyte monitoring within 24 hours of IVF initiation.

- Discontinue **IVF AS EARLY AS POSSIBLE** if patient can take adequate enteral fluids

DEFICIT THERAPY

Dehydration is significant depletion of body water and to varying degrees, electrolytes. It remains a major cause of morbidity and mortality

Severity of Dehydration:-
- Mild dehydration: No hemodynamic changes (About 5% of body weight in infants and 3% in adolescent)
- Moderate dehydration: Tachycardia (about 10% body weight in infants and 5 to6 % in adolescents)
- Severe dehydration: Hypotension with impaired perfusion (about 15% body wt in infant

Type of dehydration:-
Isonatremic serum Na 135 -145
Hyponatremic serum Na < 135
Hypernatremic serum Na > 145

Rehydration Phase in hypo/isonatremic dehydration

Calculate deficit according to degree of hydration

		Fluid volume	Type of fluid
Phase1	Resuscitation	20ml/kg can give up to 60ml/kg Each bolus 20 to 30 minutes	Normal Saline
Phase 2	First 8 hrs Calculate remaining deficit Total defcit volume – total fluid given in resuscitation phase	1/2remaining deficit+1/3 maintenance	Normal saline with 5% dextrose Add KCl Acc to potassium level
Phase 3	Next 16 hours	1/2remaining deficit+2/3 maintenance	Normal saline with 5% dextrose Add KCl Acc to potassium level

Note: Fluid Serum sodium should be checked frequently (every 4-6 hr) in case of hyponatremia and it should be corrected by no more than 10meq/L/d. In case of isonatremic dehydration, there is no need to monitor electrolyte unless patient requiring IVF more than 24 hours or having frequents diarrhea /vomiting.

Rehydration Phase in hypernatremic dehydration
Fluid deficit should be corrected for 48 hours due to hyperosmolar state of circulatory blood, Serum sodium should be checked frequently (every 2-4 hr) and it should be corrected by no more than 10meq/L/d.

Ongoing losses
Volume of ongoing losses should be measure directly (e.g. nasogstric tube, stool measurement) or estimated (e.g. 10ml/kg per diarrhea stool). It can be replaced by either oral/NG ORS or intravenous fluid in case if unable to tolerate enterally.

References:
1- Fluid and Electrolyte Administration in Children, Clinical Practice Guideline 2019, The Hospital for Sick Children (sickkids).
2- Leonard G Feld, Daniel R. Clinical Practice Guideline: Maintenance Intravenous Fluid in children, American Academy of Pediatrics: vol 14: number 6, Dec 2018.
3- Consenses Guideline for IV fluid Management: Northern California Pediatric Hospital Medicine Consortium, Originated 6/2015. Last edition 4/2019.

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